Answer Consistency under Quantization: A Controlled Study with Exact Input Transformations

Working research draft
Author and affiliation to be confirmed

8 November 2026

Abstract

Weight quantization can change how a language model answers equivalent presentations of the same problem. We study this behavior with executable answer oracles and explicit transport of displayed options to canonical answer identities. A two-model pilot illustrates why agreement alone is insufficient: four-bit SmolLM2 becomes more consistent while losing accuracy. We compare scale-only teacher distillation with a paired invariance objective, holding input exposure and optimization updates fixed, with three fit seeds and 1,152 independent evaluation bases per model. Paired calibration reduces consistency on held-out single transformations for both models. Qwen's joint correctness improves by 2.04–11.50 percentage points, while Smol's changes range from −15.62 to +4.25 points. On compositions withheld from both fitting and development selection, Qwen's consistency rises by 14.71–20.62 points, but Smol's changes range from −5.25 to +9.11. The effects depend on model and fit seed; they do not support a general preservation benefit. The released numerical RTN study separates useful agreement, wrong agreement, and fidelity to the full-precision teacher.

1 Introduction

Two answers can agree because a model follows the same reasoning, because it repeats the same mistake, or because it favors a displayed option. Quantization can change all three behaviors. Average task accuracy does not identify which individual predictions changed, while consistency by itself does not identify correct reasoning. A useful evaluation must retain the identity of an answer across transformations and distinguish jointly correct pairs from pairs that consistently select a wrong answer.

This distinction is particularly concrete in multiple-choice tasks. Permuting choices changes the correct displayed letter while preserving the underlying answer. Renaming entities or reordering independent facts should preserve that answer as well. Comparing raw letters across an option permutation would therefore measure the wrong relation. We instead store an explicit display-to-canonical bijection and recompute every correctness label from an executable oracle.

Behavior-aware quantization is established. FidelityQ targets predictive fidelity, DPQ targets uncertainty preservation, and EAQuant targets answer-supporting evidence [4, 5, 6]. KernelQuant already preserves relations among calibration representations [7]. Our study consequently does not claim that preserving behavior or relations during quantization is new. We test a narrower question: does a loss based on known semantic input correspondences improve held-out correspondence beyond giving a quantized model exactly the same transformed inputs without that paired loss?

The study provides an exact procedural evaluation with separated calibration, development, pilot, and final bases, and a matched comparison of five scale-calibration objectives. It evaluates accuracy, teacher prediction fidelity, joint correctness, and consistently wrong agreement alongside consistency. These controls make a negative outcome informative: generic calibration, data augmentation, or a trivial option preference may account for an apparent invariance gain.

2 Related work and scope of novelty

Consistency regularization across augmented inputs is established in both language and vision [1]. AugMix explicitly uses Jensen–Shannon divergence across original and augmented predictions [2]. Neither the divergence nor invariance training is introduced here. Our comparison applies these ingredients to frozen-code quantization scales with exact option transport and a controlled information boundary.

Output matching and scale optimization are prior methods rather than contributions of this study. OSTQuant uses learned transformations and teacher supervision [8]; RoleQ includes next-token distillation of scales with frozen integer codes [9]. FPTQuant studies internal function-preserving transformations [10], a different use of symmetry from transforming the question presented to a model. Prompt-sensitivity changes under quantization are also known [11]. A separate audit studies agreement in stochastic model panels [12]; our relation compares explicitly mapped presentations of one base problem. Exact transformation testing is not introduced here as a new general testing methodology. Correctness Agreement also measures the intersection of correct decisions between full-precision and quantized models on identical inputs [3]. Joint correctness and the observation that equal accuracy can hide different decisions are established; our pairs instead compare distinct mapped presentations under the same model.

KernelQuant is an especially relevant alternative: its calibration preserves cosine kernels among representations [7]. Our kernel-loss control uses the same end-to-end optimization budget as the other controls, but it is an adaptation at the final hidden representation, not a reproduction of KernelQuant's layerwise scale search, error compensation, rotation, and activation clipping. The bounded literature search found no direct match for the exact semantic-pair calibration and held-out evaluation contract used here. This is a statement about the searched sources, not proof of priority.

3 Exact tasks and correspondence

We generate arithmetic evaluation, Boolean-assignment selection, and directed acyclic graph questions. Each question has four distinct options and one correct canonical option, verified by integer arithmetic, exhaustive Boolean evaluation, or directed breadth-first search. Arithmetic expressions use addition, subtraction, and multiplication with explicit parentheses. Boolean questions ask which complete assignment makes an expression true. Graph questions ask for a reachable candidate or one at an exact shortest distance.

The transformations are a seeded nonidentity option permutation, a consistent entity renaming, and, for Boolean/graph questions, reordering independent assignment facts or directed edges. Arithmetic order is not changed. For final evaluation only, we also compose option permutation with renaming, and with renaming and fact order where supported. Compositions reuse the exact component maps rather than sampling new ones.

Calibration and development each contain 128 bases per task family; the pilot contains 64, the final split 256, and a separate structural holdout 128. Variants of a base always remain in the same split. Exact normalized structures are disjoint across splits, excluding distractor choices and arithmetic character names from the identity key. This boundary does not deduplicate arbitrary Boolean logical equivalence or graph isomorphism. Deterministic rejection resolves duplicate candidates, and the manifest retains candidate counts and rejection reasons.

The structural holdout uses five or six arithmetic operations rather than two or three, five Boolean assignment variables rather than three or four with a maximum expression depth of three rather than two, and eight to ten graph nodes rather than five to seven. A Boolean expression need not use every assignment variable, so this split does not guarantee a strict increase in logical difficulty for every question.

For base i, presentation t, and displayed option j, let $m_{it}(j)$ denote the canonical option identity. With the official chat template, the model produces next-token option logits z_{itj} for the

verified single tokens A–D. We predict

$$\widehat{a}_{it} = m_{it}\left(\arg\max_j z_{itj}\right).$$

Probabilities are computed in FP32; predictions use logits directly, avoiding artificial ties from BF16 softmax rounding. For paired losses, the four-option probability vector is transported by $p^{\mathrm{can}}_{it,m_{it}(j)} = \mathrm{softmax}(z_{it})_j$. This is constrained next-token scoring, not evaluation of free generation.

Let Y_{it} indicate correctness under the exact oracle. For each non-original presentation define

$$C_{it} = \mathbf{1}[\widehat{a}_{i0} = \widehat{a}_{it}], \qquad J_{it} = Y_{i0}Y_{it}, \qquad W_{it} = C_{it}(1 - Y_{i0}).$$

Here C is consistency, J joint correctness, and W consistently wrong agreement. A unique correct canonical option makes J a subset of C. We first average over the available variants of each base, then over bases. The primary held-out endpoint uses entity renaming and fact order only; compositions and structural holdout are reported separately. Pointwise teacher fidelity is the fraction of canonical predictions matching BF16 on the same input.

Consistency change necessarily involves pointwise prediction change. If F_{i0} and F_{it} indicate BF16-to-quantized answer flips, then $|C^q_{it} - C^f_{it}| \le F_{i0} + F_{it}$. We use this elementary bound as an implementation diagnostic, not a novel theorem or an independent failure mechanism.

4 Quantization and matched calibration

The reference models use BF16. Every transformer linear weight except the embedding and output head is quantized independently within input-column groups of 128, including tail groups. For bit width b, the RTN scale is $s_g = \max_{k\in g} |w_k|/(2^{b-1} - 1)$ and codes are

$$q_k = \mathrm{clip}\Big(\mathrm{round}(w_k/s_g), -2^{b-1}, 2^{b-1} - 1\Big).$$

All-zero groups reconstruct exactly zero. The pilot evaluates W8, W4, and W3. The calibration comparison uses W4. Integer codes, pretrained weights, biases, embeddings, output head, and normalization gains remain fixed. Only positive group scales are learned:

$$\widetilde{s}_g = s_g \exp(\delta_g), \qquad \delta_g = 0 \text{ initially.}$$

This stable relative parameterization exactly reproduces RTN at initialization. Weights are reconstructed and rounded to BF16 during forward passes. The experiment establishes numerical weight-quantization behavior, not packed-format deployment parity, memory savings, or inference acceleration.

4.1 Objectives and supervision

The teacher cache contains complete final next-token vocabulary logits and the final normalized hidden vector for each calibration original and its option-permuted partner. No gold answer labels are training targets. All optimized methods share the temperature-one loss $\mathcal{L}_{\mathrm{KD}} = \mathrm{mean}_x \mathrm{KL}(P_f(\cdot \mid x) \| P_q(\cdot \mid x))$ over the full vocabulary. They differ only in input presentation or the extra regularizer:

1. Original-only KD duplicates each original, keeping eight prompt rows per update.
2. Augmented KD uses each original and its option-permuted partner independently.
3. Paired calibration adds λ times the mean Jensen–Shannon divergence between the canonically transported four-option distributions of each true pair.

4. Random-pair calibration adds the same loss after shifting the variants cyclically to different bases in the shuffled minibatch.

5. The kernel control adds $\lambda B^{-2}\|H_f H_f^\top - H_q H_q^\top\|_F^2$, where H contains the unit-normalized final hidden vectors of the same $B = 8$ prompt rows, with $H \in \mathbb{R}^{B\times d}$ and model hidden width d.

The paired objective imposes extra invariance. An inconsistent teacher can therefore disagree with its regularizer; this is not exact preservation of every teacher relation. Accuracy, fidelity, and consistently wrong agreement are required to interpret any consistency gain.

Every scale-calibration fit starts from identical RTN scales and uses AdamW with learning rate 0.001, betas $(0.9, 0.999)$, no weight decay, gradient-norm clipping at one, and zero dropout. One shuffled epoch consists of 96 updates with four bases and eight prompts per update, covering 384 bases and 768 prompts. All loss computations and scale parameters are FP32, with non-reentrant gradient checkpointing and explicit rejection of nonfinite losses/gradients. We record actual nonpadding and padded token counts, batch order, trainable parameter count, elapsed optimization time, and peak allocated optimization memory. Original-only KD may differ in token exposure; the primary paired and augmented comparison uses exactly the same input rows and order.

4.2 Selection and independent evaluation

After inspecting the pilot and Qwen development, but before any final model inference, we added a secondary global-label diagnostic. For each calibration input let $d_x = z_f(x) - z_{\mathrm{W4}}(x)$ be the four displayed-option logit differences. The fixed offsets are $b = \mathrm{mean}_x[d_x - \mathrm{mean}_j(d_{xj})\mathbf{1}]$, using calibration originals and option permutations only. At evaluation we predict from $z_{\mathrm{W4}} + b$. This rule fits four offsets without oracle targets, an optimizer, development selection, or encoder updates. It tests a global letter-preference explanation and has a different budget from the five matched scale objectives; it is not the primary comparison.

For each regularized objective we search $\lambda \in \{0.1, 1, 10\}$ using fit seed 17 and development data only. Candidates within one percentage point of augmented-KD overall development accuracy are ranked by held-out joint correctness, then held-out consistency, then smaller λ. If none meets the accuracy constraint, ranking first uses overall accuracy and then the same criteria. Every candidate outcome is retained. Selected seed-17 fits are reused, and the chosen setting is fitted with seeds 29 and 43. Intermediate checkpoints are not selected. Renaming and fact order are thus absent from gradient updates but present in development selection. Only compositions are withheld from both stages. Our transfer claim concerns fresh bases and unseen compositions, not primitive transformation families unseen to the complete selection procedure.

Both models' choices are frozen before final model inference. Primary comparisons are paired versus augmented KD on held-out single transformations, within each model and fit seed. Percentile 95% intervals use 1,000 paired resamples of whole bases, seed 4108, carrying all variants together. Fit seeds are separate fits, not extra independent evaluation problems. Intervals are conditional on each fitted model and are not adjusted for the multiple reported endpoints and subgroup comparisons. The report includes all task, split, and transformation outcomes. A fixed displayed-A rule is an untrained descriptive negative control. Optimization time is also compared; an overhead exceeding 10% requires a time-matched augmented-KD control before claiming a matched-compute gain.

5 Pilot observations

The exploratory pilot evaluated 192 bases, 704 inputs, and 512 original-variant pairs per model. Qwen W4 broke 64 previously consistent pairs and repaired 63, for a base-macro consistency difference of -0.35 percentage points (95% interval $[-5.38, 4.51]$). W3 reduced consistency by

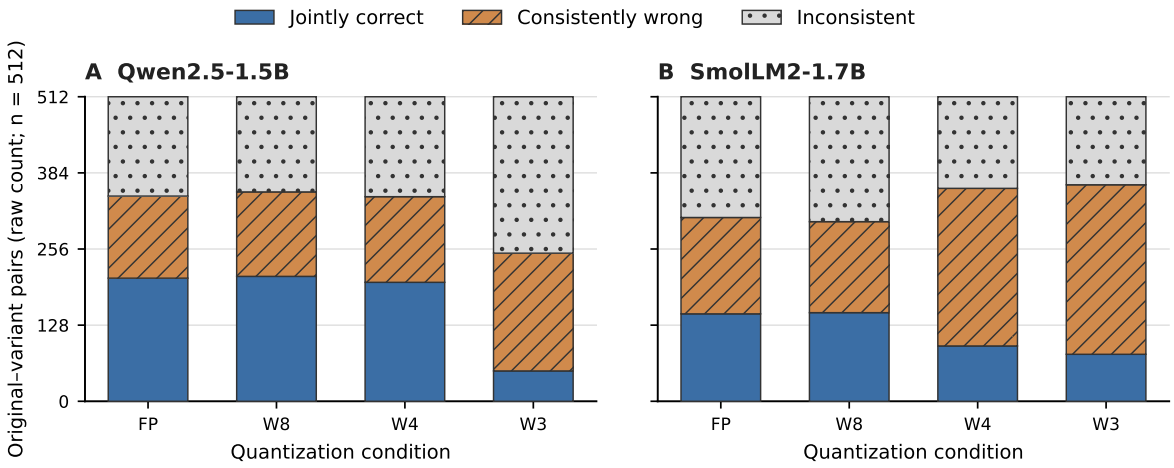


Figure 1: Exploratory pilot outcomes for all 512 original-variant pairs per model. Solid segments are jointly correct, hatched segments agree on a wrong canonical option, and dotted segments disagree. These are descriptive raw pair counts, not equal-base-weighted rates. BF16 is the reference; quantized weights are dequantized for BF16 arithmetic.

19.44 points (interval $[-25.53, -13.45]$). The predefined rule of at least ten broken pairs triggered calibration; this is a development rule, not a significance test.

Smol provides a counterexample to treating consistency as quality. W4 reduced its base-macro accuracy from 40.02% to 26.52%, while consistency rose from 60.33% to 68.92%. W3 selected displayed A on every pilot input. Figure 1 separates consistency into jointly correct and consistently wrong pairs. These pilot findings motivated the displayed-A control before final outcomes were observed; they do not establish a benefit of the calibrated method.

6 Independent final results

The independent evaluation contains 768 final bases and 384 structural-holdout bases, rendered into 6,144 inputs per condition. Each model has 21 conditions: BF16 and three RTN precisions, fifteen selected scale fits, the displayed-A rule, and the calibration-only four-logit correction. No final outcome is used for model, loss, checkpoint, or hyperparameter selection. Unless a split is named, the following differences pool the 1,152 independent bases with equal base weight. Reported seed ranges describe three individual fits; they are not confidence intervals over training randomness. Table 1 lists the pooled point estimates; the released comparison files retain the corresponding base-bootstrap intervals for the pooled data, each split, and each task family.

6.1 Qwen: correctness gains without primary consistency gains

On held-out single transformations, paired calibration changes consistency by -1.87, -1.87, and -4.51 percentage points versus same-seed augmented KD (seeds 17, 29, and 43). The corresponding joint-correctness differences are $+11.50$, $+3.47$, and $+2.04$ points, with base-bootstrap 95% intervals $[8.55, 14.37]$, $[0.82, 6.08]$, and $[-0.52, 4.77]$. Consistently wrong agreement falls by 5.34–13.37 points. Thus the primary consistency outcome does not improve: more useful agreement partly replaces wrong agreement, while total agreement is lower.

Compositions are a distinct secondary endpoint withheld from both updates and development selection. Here consistency rises by 14.71–20.62 points and joint correctness by 10.63–16.19 points. Every corresponding per-fit interval lies above zero. However, consistently wrong composition agreement also rises by 4.08 and 9.42 points at seeds 29 and 43; seed 17 changes it by -0.17

Table 1: Paired-minus-augmented differences in percentage points over 1,152 bases per model. Singles denote held-out entity renaming/fact order; compositions were withheld from fitting and selection. C is consistency, J joint correctness, and W consistently wrong agreement. Seeds are distinct fits evaluated on the same bases.

		Singles			Compositions	
Model	Seed	ΔC	ΔJ	ΔW	ΔC	ΔJ
Qwen	17	−1.87	+11.50	−13.37	+16.02	+16.19
Qwen	29	−1.87	+3.47	−5.34	+14.71	+10.63
Qwen	43	−4.51	+2.04	−6.55	+20.62	+11.20
Smol	17	−8.77	+4.25	−13.02	+9.11	+9.81
Smol	29	−31.94	−15.62	−16.32	−5.25	−5.64
Smol	43	−13.06	−1.65	−11.41	+8.16	+4.56

points with an interval spanning zero. This is transfer of an invariance objective with a mixed error profile. It is not uniformly beneficial consistency preservation.

The fidelity endpoint makes the distinction sharper. Relative to augmented KD, paired calibration reduces held-out-single agreement with BF16 by 6.42–13.11 points, and composition agreement with BF16 by 3.39–12.63 points. The method can improve oracle correctness while moving away from the full-precision teacher. These outcomes contradict an unqualified claim that the paired loss preserves the teacher's behavior.

6.2 Smol: the Qwen pattern does not transfer reliably

Smol's held-out-single consistency falls by 8.77–31.94 points under paired calibration. Joint correctness changes by +4.25, −15.62, and −1.65 points at seeds 17, 29, and 43. Consistently wrong agreement falls by 11.41–16.32 points, but this reduction does not ensure useful replacement: at seed 29, correct agreement also falls substantially. Composition consistency changes by +9.11, −5.25, and +8.16 points; composition joint correctness changes by +9.81, −5.64, and +4.56. Thus Qwen's positive composition result does not establish a benefit across the two models and three fits. All fits completed the fixed 96 updates with finite losses; no final outcome was used to replace the selected strength. Across the three fits, held-out-single teacher fidelity falls by 12.72–33.20 points relative to augmented KD, and composition teacher fidelity falls by 17.71–38.15 points. The loss is therefore not a reliable teacher-preservation objective in either model.

The uncalibrated three-bit Smol model selects displayed A on 6,139 of 6,144 independent evaluation inputs (99.92%), extending the pilot's all-A behavior without claiming exact identity with a constant rule on the final data. Figures 2–4 show the 768-base final split separately; the pooled contrasts above also include the 384 structural-holdout bases, whose results remain separately available.

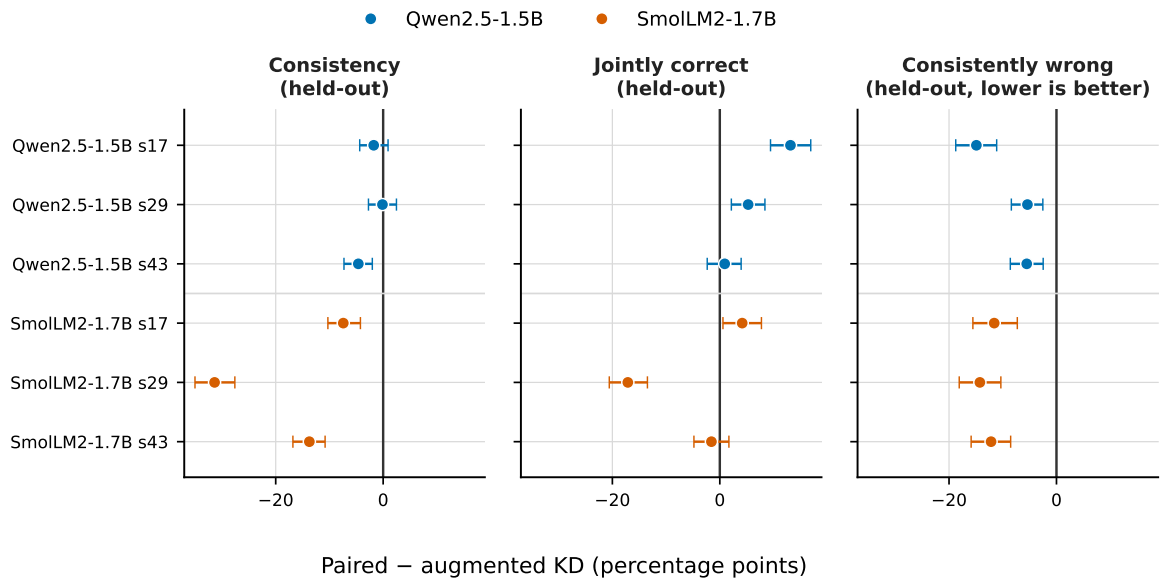


Figure 2: Primary paired-minus-augmented contrasts on held-out single transformations in the final split (768 bases per model). Each row is one fit seed; horizontal intervals are percentile 95% intervals from 1,000 paired base resamples. The three panels distinguish total agreement, joint correctness, and consistently wrong agreement. Lower is better only for the last panel. Intervals are conditional on each fit and unadjusted.

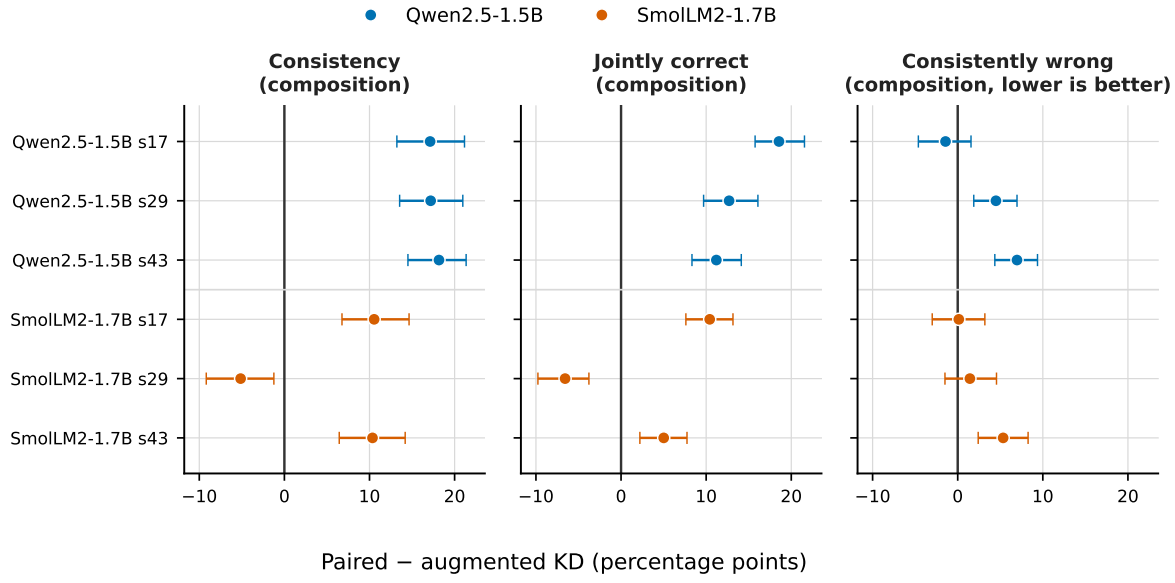


Figure 3: Secondary contrasts on final-split compositions withheld from both gradient updates and development selection (768 bases per model). Conventions match Figure 2. A consistency gain may contain both more jointly correct and more consistently wrong pairs; these components are shown separately.

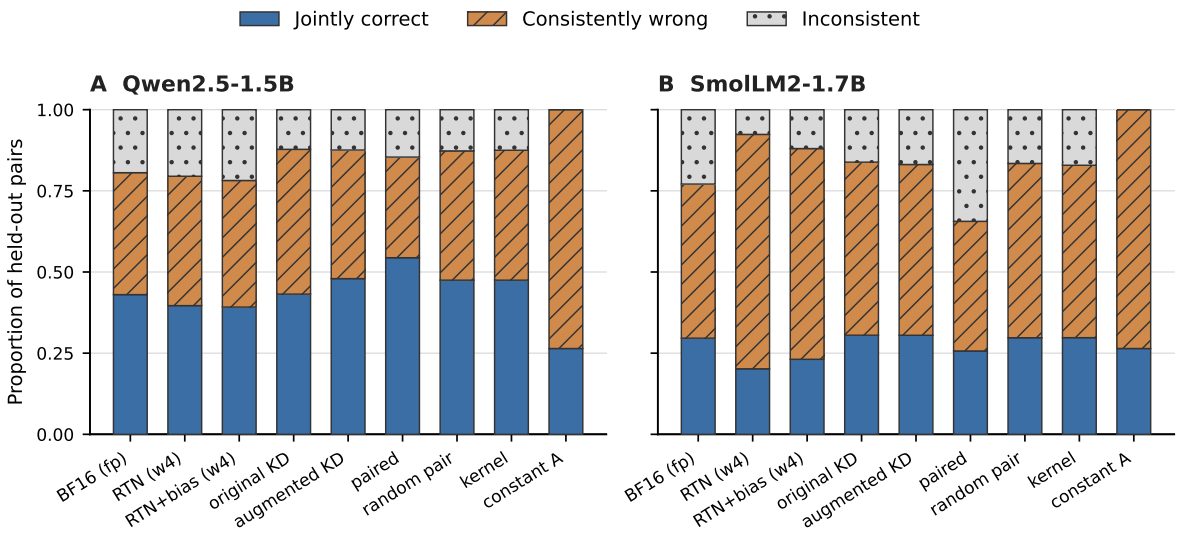


Figure 4: Absolute held-out-single outcomes on the final split. Each base receives equal weight; learned conditions are descriptive means over seeds 17, 29, and 43, while untrained conditions have one result. Solid segments are jointly correct, hatched segments consistently wrong, and dotted segments disagree. The kernel condition is the matched cosine-Gram adaptation; RTN+bias is the separate calibration-only four-logit rule. The constant-A rule demonstrates that perfect consistency need not imply useful answers.

7 Limitations and reproducibility

This is a controlled study of two small instruction models on English procedural multiple-choice problems. It does not establish performance on natural-language paraphrases, free generation, long reasoning traces, deployment kernels, or larger contemporary models. A higher consistency score can result from a less informative predictor, including the displayed-A rule. The full-precision teacher is also not an oracle, and enforcing invariance can trade teacher fidelity against correctness.

The comparison isolates objectives within a signed RTN initialization and a single-epoch scale schedule. We do not reproduce full KernelQuant or benchmark against deployed AWQ/GPTQ pipelines. Their absence limits claims about competitive quantization quality. The kernel adaptation is an objective control, not a substitute for those complete systems. The three-strength search does not equalize regularizer magnitudes, and an individual seed's development results select the strength for all three fits. This design measures repeatability of that selected recipe rather than robustness under a multiseed selection procedure. An unquantized scale-optimization control is absent. We therefore cannot attribute an observed calibration benefit uniquely to compression or exclude generic invariance fine-tuning as an explanation. The four-logit diagnostic tests one fixed mean-drift correction, not every possible position-bias explanation. Even an improvement over random pairing does not establish a particular internal reasoning mechanism. Equivalent Boolean functions or isomorphic graphs can cross the declared exact-structure split boundary. The confidence intervals describe variation across generated bases conditional on the fitted models and the fixed prompt/format; they do not establish broad task-population guarantees.

The protocol, amendments, exact oracle structures, option maps, configurations, per-input predictions, all development candidates, fit/timing records, and plotting code accompany the reproducibility package. Upstream pretrained weights and full-vocabulary teacher caches are not redistributed. The released scale-calibration code and pinned original model revisions allow fits to be regenerated. The draft separates the exposed pilot from the independently reserved final split.

8 Executed settings

Table 2 records the common design. Model-specific strengths were fixed from development before final inference. Both models completed all 21 final conditions. Saved predictions reproduce all condition and subgroup summaries and paired intervals in CPU replay; the offline suite passes 106 tests.

The pinned model revisions are:

Qwen: `989aa7980e4cf806f80c7fef2b1adb7bc71aa306`
Smol: `31b70e2e869a7173562077fd711b654946d38674`

Table 2: Study settings and authoritative evidence

Item	Value	Evidence
Models	Qwen2.5-1.5B-Instruct; SmolLM2-1.7B-Instruct	Pinned configs
Arithmetic	Two/three operations; five/six in structural holdout	Generator
Boolean	Three/four assignment variables; five in structural holdout	Generator
Graphs	DAGs with five–seven nodes; eight–ten in structural holdout	Generator
Bases per family	Calibration 128; development 128; pilot 64; final 256; structural holdout 128	Protocol and data manifests
Split seeds	1108, 2108, 3108, 4108, 5108, respectively	Pinned configs
Quantizer	Signed grouped RTN; 128 input columns per row/group; W4 calibration; BF16 forward arithmetic	Run manifests
Optimization	AdamW; 0.001 learning rate; 96 updates; four bases/eight prompts per update	Fit manifests
Fit seeds	17, 29, 43	Fit manifests
Lambda candidates	0.1, 1, 10; development-only selection	Frozen selections
Selected Qwen strengths	Paired 10; random pair 1; kernel 0.1	Qwen selection JSON
Selected Smol strengths	Paired 10; random pair 0.1; kernel 10	Smol selection JSON
Trainable scale deltas	Qwen 10,235,904; Smol 12,582,912	Fit manifests
Scoring	Official chat template; next-token A–D; batch 8; maximum 1,024 prompt tokens without truncation	Scorer/configs
Intervals	1,000 paired base resamples; seed 4108	Comparison JSON
Hardware/runtime	Radeon RX 9070 XT; ROCm PyTorch 2.12.0+rocm7.14.0; Transformers 4.57.1; Python 3.12	Run manifests

Tool-use disclosure

DeepSeek V4.1 Flash implemented the experimental code under Codex supervision. Codex assisted with design, literature checking, execution, tests, and drafting. The manuscript is a research draft rather than a record of peer review.

References

[1] Qizhe Xie, Zihang Dai, Eduard Hovy, Minh-Thang Luong, and Quoc V. Le. Unsupervised Data Augmentation for Consistency Training. NeurIPS, 2020. https://arxiv.org/abs/1904.12848

[2] Dan Hendrycks et al. AugMix: A Simple Data Processing Method to Improve Robustness and Uncertainty. arXiv:1912.02781v2, 2020. https://arxiv.org/html/1912.02781v2

[3] Baha Rababah et al. The Illusion of Equivalency: Statistical Characterization of Quantization Effects in LLMs. arXiv:2607.08734v2, 2026. https://arxiv.org/html/2607.08734v2

[4] Chen et al. FidelityQ: Preserving Predictive Fidelity in Ultra-Low-Bit Language Model Quantization. arXiv:2610.14934, 2026. https://arxiv.org/html/2610.14934

[5] Yang et al. Target-Aware Calibration Data Selection for Preserving Uncertainty in Quantized Language Models. Findings of EMNLP, 2026. https://aclanthology.org/2026.findings-emnlp.670/

[6] Yeji Kim, Mi-Young Kim, and Randy Goebel. When Quantization Preserves Accuracy but Not Evidence: Explanation-Aware Post-Training Quantization for Medical LLMs. EMNLP, pp. 41629–41651, 2026. https://aclanthology.org/2026.emnlp-main.1828/

[7] Park et al. KernelQuant: Post-Training Quantization via Representation Kernel Preservation. EMNLP, pp. 19833–19850, 2026. https://aclanthology.org/2026.emnlp-main.856/

[8] Hu et al. OSTQuant: Refining Large Language Model Quantization with Orthogonal and Scaling Transformations for Better Distribution Fitting. arXiv:2501.13987, 2025. https://arxiv.org/html/2501.13987

[9] Johannes Knittel, Alec Helbling, and Hanspeter Pfister. RoleQ: Role-Aligned Scales for Converting Language Models to Ternary. arXiv:2610.21808, 2026. https://arxiv.org/html/2610.21808

[10] van Breugel et al. FPTQuant: Function-Preserving Transforms for LLM Quantization. arXiv:2506.04985v2; ICML, 2026. https://arxiv.org/abs/2506.04985

[11] Fu et al. Quantized but Deceptive? A Multi-Dimensional Truthfulness Evaluation of Quantized LLMs. EMNLP, pp. 30435–30458, 2025. https://aclanthology.org/2025.emnlp-main.1548/

[12] Ananto Nayan Bala. Does Quantization Change What Agreement Means? A Paired Audit of Homogeneous LLM Panels. UncertaiNLP, 2026. https://aclanthology.org/2026.uncertainlp-main.37/